B.Tech. (Computer Engg.) V^th Semester Examination, 2023
Computer Networks
Paper No. CEN-505

Time: Three Hours **Maximum Marks: 60**

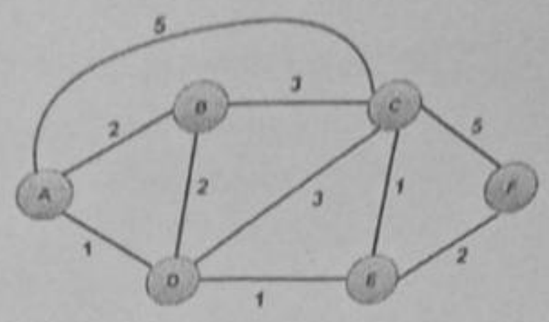
Write your roll no. immediately on receipt of this question paper
Note: Attempt all question. All questions carry equal marks. Assume suitable missing data, if any.

Q.No./ CO's No.	Content of Questions	Marks
1. (a)/ CO1	Compare and contrast OSI/ISO and TCP/IP Reference model with respect to data format and protocols used in each layer.	6
1. (b)/ CO1	Find the multiplexer output sequence if six stations A,B,C, D, E and F are sending the 1,0, 0,1, silent ,1 in CDMA	6
2. (a)/ CO2	Here is a network having bandwidth of 1MBps. A message of size 1000 bytes has to be sent. The packet switching technique is used. Each packet contains a header of 100 bytes. On the following, in how many packets the message must be divided so that total time taken is minimum? (propagation delay is ignored here).  i. 1 Packet ii. 5 Packets iii. 10 Packets iv. 20 Packets Calculate Total time taken for each case.	6
2. (b)/ CO2	Consider a packet of length 1,500 bytes which begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches, the propagation speed on all the links is $2.5*10^8$ m/s transmission rates of all three links are 2 Mbps, the packet switch processing delay is 3msec, the length of the first link is 5,000 km, the length of the second link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is end to end delay?	6
3. (a)/ CO3	Consider a 128×10^3 bits/sec satellite communication link with one-way propagation delay of 150 msec. Selective Retransmission (repeat) protocol is used on this link to send data with a frame size of 1 KB. Neglect the transmission time of acknowledgement. What is the minimum number of bits required for the sequence number field to achieve 100% utilization?	6

3. (b)/ CO3	A sender needs to send the four data items Ox4365, OxAB1D, Ox02BC, and Ox35FE. a. Find the checksum at the sender site. b. Find the checksum at the receiver site if there is no error. c. Find the checksum at the receiver site if the third data item is changed to Ox02DC.	6

OR

3'. (a)/ CO3	Compare and contrast the following: i. Stop and Wait protocol ii. Go back N protocol iii. Selective repeat protocol	6
3'. (b)/ CO3	Host A is sending data to host B over a full duplex link. A and B are using the sliding window protocol for flow control. The send and receive window sizes are 5 packets each. Data packets (sent only from A to B) are all 1000 bytes long and the transmission time for such a packet is 50 ms. Acknowledgment packets (sent only from B to A), are very small and require negligible transmission time. The propagation delay over the link is 200 ms. What is the maximum achievable throughput in this communication?	6
4. (a)/ CO4	Sixteen stations, numbered 1 through 16, are contending for the use of a shared channel by using the adaptive tree walk protocol. If all the stations whose addresses are prime numbers suddenly become ready at once, how many bit slots are needed to resolve the contention?	6
4. (b)/ CO4	Explain the working principle of CSMA/CA in detail. What is the problem in CSMA/CD that CSMA/CA is trying to resolve? Justify your answer with suitable example.	6
5. (a)/ CO5	For a single network having IP Address 192.20.10.0. Divide this network into three subnets such that the second subnet is divided into two equal subnets. Find the following for each subnet: i. Network address ii. Total number of IP Addresses iii. Total number of hosts that can be configured iv. Range of IP Addresses v. Direct Broadcast Address	6
5. (b)/ CO5	Create local routing table for A to F and full routing table of router A and C for the following given subnet using Link state routing. Enter the values in the routing table step by step using Dijkstra Algorithm.	6



OR

5'. (a)/ CO5 An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows: **6**

i. The first group has 64 customers; each needs approximately 256 addresses.
ii. The second group has 128 customers; each needs approximately 128 addresses.
iii. The third group has 128 customers; each needs approximately 64 addresses.

Design the sub-blocks and find out how many addresses are still available after these allocations.

5'. (b)/ CO5 Consider the following subnet. Distance vector routing is used, and the following vectors have just come in to router F: from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4). The measured delays to B, D, and E, are 9, 6, and 3, respectively. What is F's new routing table? Give both the outgoing line to use and the expected delay. **6**

